Cisco Networking Lab: DHCP Snooping and Dynamic ARP Inspection (DAI)

1. Objective

The objective of this lab is to secure a Layer 2 network against:

- Rogue DHCP Server attacks.
- ARP spoofing and ARP poisoning.
- Unauthorized DHCP server messages.
- Invalid ARP packets.

We use DHCP Snooping to identify legitimate DHCP bindings and Dynamic ARP Inspection (DAI) to validate ARP packets.

2. Tools and Devices

- Cisco Packet Tracer
- 1 × Cisco Catalyst 2960 Switch
- 3 × PCs
- 2 × Servers

3. Network Topology

Legitimate DHCP Server — Fa0/1

Rogue DHCP Server — Fa0/2

PC0 — Fa0/3

PC1 — Fa0/4

PC2 — Fa0/5

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Cisco 2960

4. IP Addressing Table

Device	IP Address	Role
Legitimate Server	192.168.1.2	Authorized DHCP Server

Rogue Server	192.168.1.3	Unauthorized DHCP Server
PC0	DHCP	Client
PC1	DHCP	Client
PC2	DHCP	Client

Subnet Mask: 255.255.255.0

Legitimate DHCP Pool

Setting	**Value**
Pool Name	LEGIT-DHCP
Default Gateway	192.168.1.1
DNS Server	8.8.8.8
Start IP	192.168.1.10
Subnet Mask	255.255.255.0
Maximum Users	50

Rogue DHCP Pool

Setting	**Value**
Pool Name	ROGUE-DHCP
Default Gateway	192.168.1.254
DNS Server	1.1.1.1
Start IP	192.168.1.100
Subnet Mask	255.255.255.0
Maximum Users	50

5. DHCP Snooping Configuration

Step 1: Enable DHCP Snooping Globally

enable

configure terminal

ip dhcp snooping

Explanation: Enables DHCP Snooping globally on the switch.

Step 2: Enable DHCP Snooping on VLAN 1

ip dhcp snooping vlan 1

Explanation: Enables DHCP Snooping for VLAN 1.

Step 3: Configure the Trusted Interface

interface fa0/1

ip dhcp snooping trust

exit

Explanation: Fa0/1 is connected to the legitimate DHCP Server. It is configured as a trusted port.

All other interfaces remain untrusted by default.

Step 4: Verify DHCP Snooping

end

show ip dhcp snooping

Expected configuration:

- DHCP Snooping: Enabled
- VLAN 1: Configured
- Fa0/1: Trusted
- Fa0/2–Fa0/5: Untrusted

6. DHCP Snooping Testing

Initially, DHCP Snooping was disabled.

Both DHCP servers were able to provide IP addresses to clients.

Before DHCP Snooping

Client Assigned IP	DHCP Source

PC0	192.168.1.10	Legitimate
PC1	192.168.1.102	Rogue
PC2	192.168.1.103	Rogue

After enabling DHCP Snooping, the clients requested fresh IP addresses.

After DHCP Snooping

Client	Assigned IP
PC0	192.168.1.12
PC1	192.168.1.13
PC2	192.168.1.14

The clients received addresses from the legitimate DHCP pool.

7. Verify DHCP Snooping Binding Table

show ip dhcp snooping binding

Observed output:

MAC Address	IP Address	VLAN	Interface
00:E0:A3:13:25	192.168.1.12	1	Fa0/3
00:D0:97:29:BC:6D	192.168.1.13	1	Fa0/4
00:90:21:69:A5	192.168.1.14	1	Fa0/5

Total number of bindings: **3**

The switch learned the IP-to-MAC mappings of the DHCP clients.

8. Dynamic ARP Inspection (DAI) Configuration

DAI validates ARP packets to prevent ARP spoofing and poisoning attacks.

It uses the DHCP Snooping binding table to validate dynamically assigned client IP and MAC addresses.

Step 1: Enable DAI on VLAN 1

enable

configure terminal

ip arp inspection vlan 1

end

Explanation: Enables Dynamic ARP Inspection on VLAN 1.

Step 2: Configure the Trusted Interface

configure terminal

interface fa0/1

ip arp inspection trust

exit

end

Explanation: Configures Fa0/1 as a trusted DAI interface because it is connected to the legitimate DHCP Server.

Client ports and the rogue server port remain untrusted by default.

Step 3: Verify DAI Interfaces

show ip arp inspection interfaces

Expected interface configuration:

Interface	Trust State
Fa0/1	Trusted
Fa0/2	Untrusted
Fa0/3	Untrusted
Fa0/4	Untrusted
Fa0/5	Untrusted

Step 4: Enable ARP Packet Validation

configure terminal

ip arp inspection validate src-mac dst-mac ip

end

Explanation:

- src-mac: Validates the source MAC address.
- dst-mac: Validates the destination MAC address.
- ip: Validates IP addresses in ARP packets.

Step 5: Verify DAI Configuration

show ip arp inspection vlan 1

Expected configuration:

- Source MAC Validation: Enabled
- Destination MAC Validation: Enabled
- IP Address Validation: Enabled
- VLAN 1 Configuration: Enabled
- Operation: Active

9. Legitimate ARP Communication Testing

PC0 IP: 192.168.1.12

PC1 IP: 192.168.1.13

From PC0, run:

ping 192.168.1.13

Result: Successful ping replies were received.

Verify ARP Table

arp -a

Observed mapping:

Internet Address Physical Address Type

192.168.1.13 00d0.9729.bc6d dynamic

This confirms that PC0 successfully learned PC1's MAC address.

10. ARP Traffic Observation

Packet Tracer Simulation Mode was used to observe ARP traffic.

Steps:

1. Switch to Simulation Mode.
2. Open Edit Filters.
3. Unselect all filters.
4. Select only ARP.
5. Clear the ARP cache on PC0:

arp -d

6. Generate traffic:

ping 192.168.1.14

7. Use Capture/Forward to observe the ARP packets.

Observation:

- ARP Request is broadcast within the VLAN.
- The intended destination responds with an ARP Reply.
- ARP Reply is normally unicast to the requesting device.

11. DAI Statistics Verification

show ip arp inspection statistics vlan 1

This command displays:

- Forwarded ARP packets
- Dropped ARP packets
- DHCP binding validation drops
- ACL drops
- Source MAC validation failures
- Destination MAC validation failures
- IP validation failures

12. Final Verification Commands

show ip dhcp snooping

show ip dhcp snooping binding

show ip arp inspection vlan 1

show ip arp inspection interfaces

show ip arp inspection statistics vlan 1

Note: Some Packet Tracer switch IOS commands may have limited support.

13. Important Concepts Learned

DHCP Snooping

DHCP Snooping protects the network against unauthorized DHCP servers.

Trusted Port

A trusted port allows DHCP server messages from an authorized DHCP server.

Untrusted Port

An untrusted port blocks unauthorized DHCP server messages.

DHCP Snooping Binding Table

Stores client IP addresses, MAC addresses, VLANs, lease times, and switch interfaces.

Dynamic ARP Inspection

DAI validates ARP packets to protect the network against ARP spoofing and poisoning.

ARP Request vs ARP Reply

- ARP Request: Broadcast
- ARP Reply: Normally Unicast

14. Interview Questions and Answers

Q1. What is DHCP Snooping?

DHCP Snooping is a Layer 2 security feature that protects a network against rogue DHCP servers.

Q2. What is a Rogue DHCP Server?

A Rogue DHCP Server is an unauthorized DHCP server that provides incorrect IP configuration to network clients.

Q3. What is a trusted port?

A trusted port allows DHCP server messages from an authorized DHCP server.

Q4. What is an untrusted port?

An untrusted port blocks unauthorized DHCP server messages.

Q5. What is a DHCP Snooping Binding Table?

It stores IP-to-MAC mappings, VLAN information, lease time, and interface information for DHCP clients.

Q6. What is Dynamic ARP Inspection?

DAI is a Layer 2 security feature that validates ARP packets to prevent ARP spoofing attacks.

Q7. How does DAI work?

DAI checks ARP packets against the DHCP Snooping binding database and drops invalid packets on untrusted interfaces.

Q8. Why is DHCP Snooping used with DAI?

DHCP Snooping creates the IP-to-MAC binding database that DAI uses for validation.

Q9. Can DAI work without DHCP Snooping?

Yes. DAI can use ARP ACLs to validate statically assigned IP-to-MAC mappings.

Q10. What is ARP Spoofing?

ARP Spoofing is an attack in which an attacker sends forged ARP messages to associate an incorrect MAC address with an IP address.

Q11. What is the difference between DHCP Snooping and DAI?

DHCP Snooping protects against rogue DHCP servers, while DAI protects against invalid or spoofed ARP packets.

Q12. What happens when an invalid ARP packet is detected?

DAI drops the invalid ARP packet on an untrusted interface.

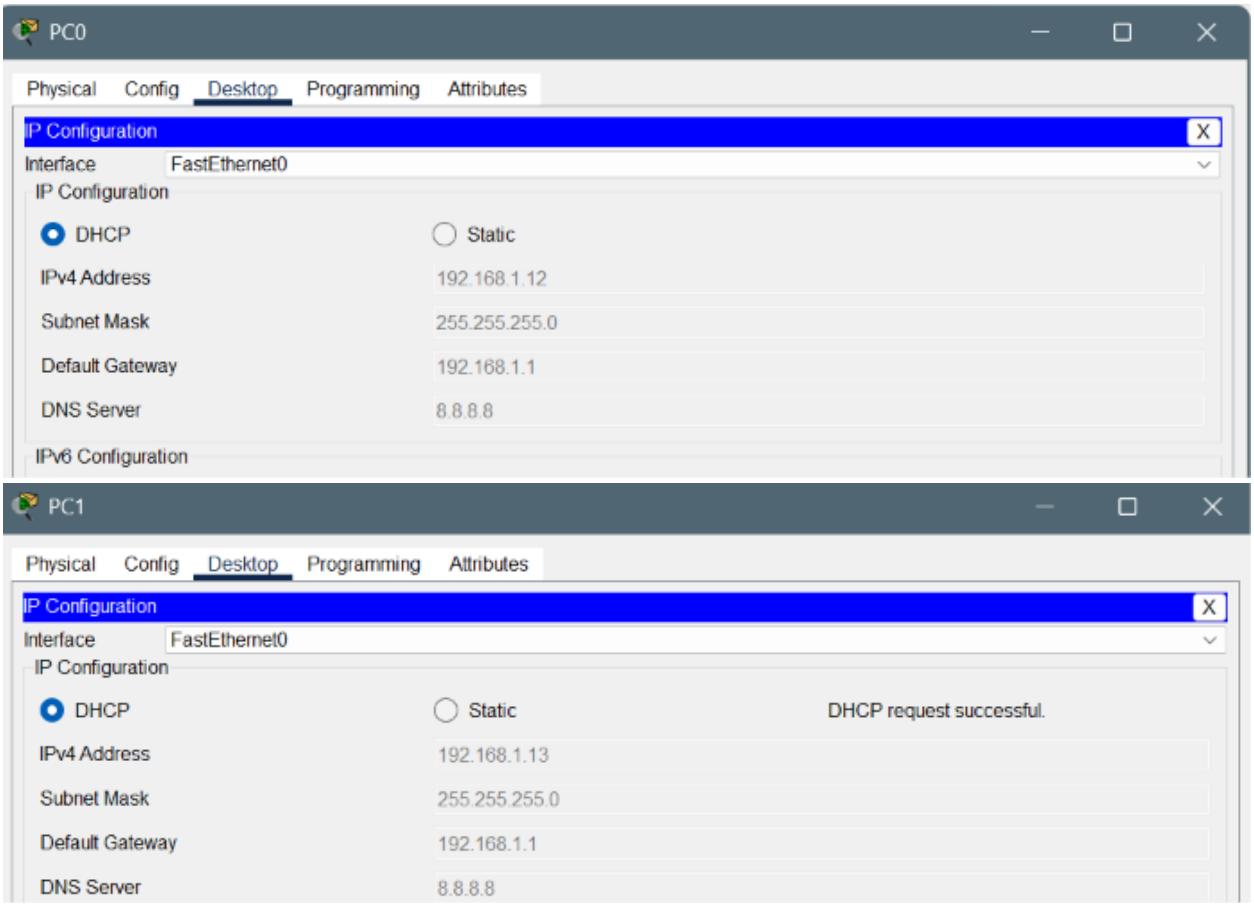
15. Conclusion

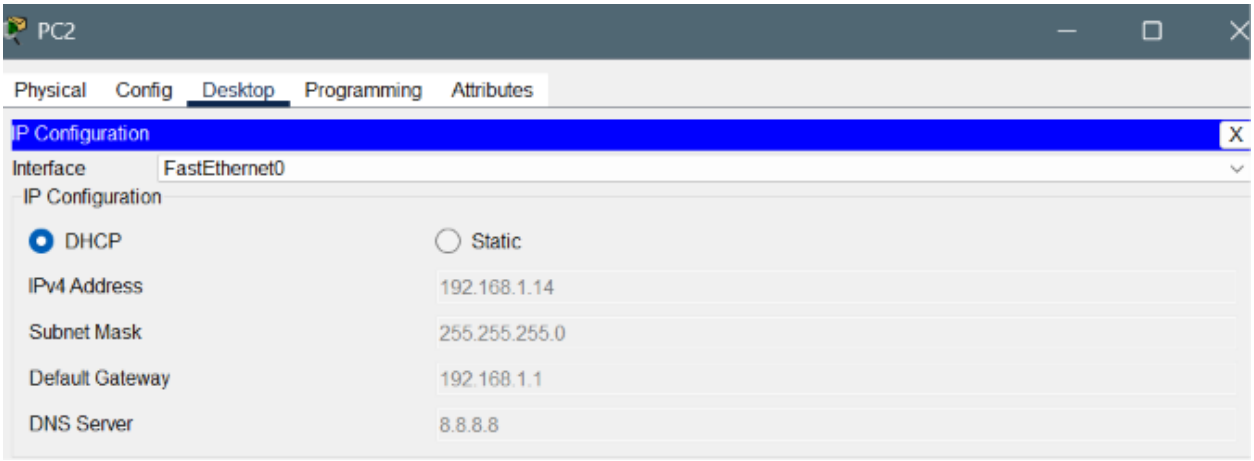
In this practical lab, DHCP Snooping and Dynamic ARP Inspection were configured on a Cisco Catalyst 2960 switch.

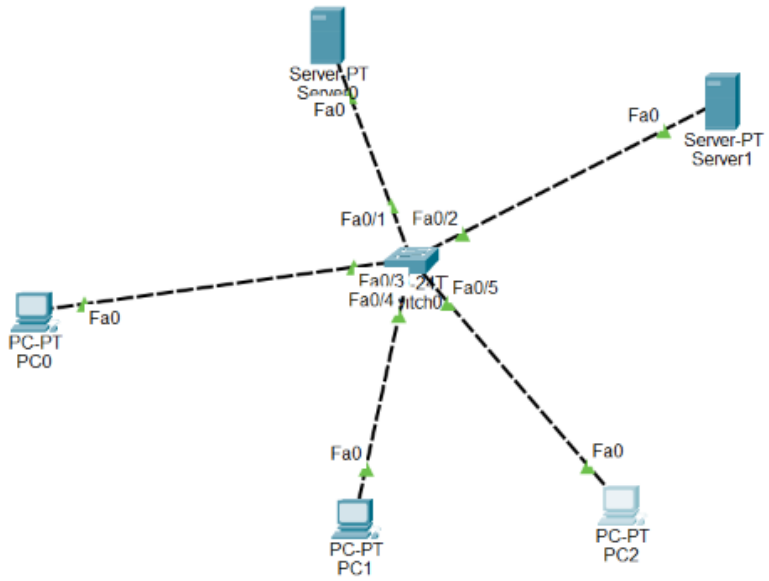
The lab demonstrated:

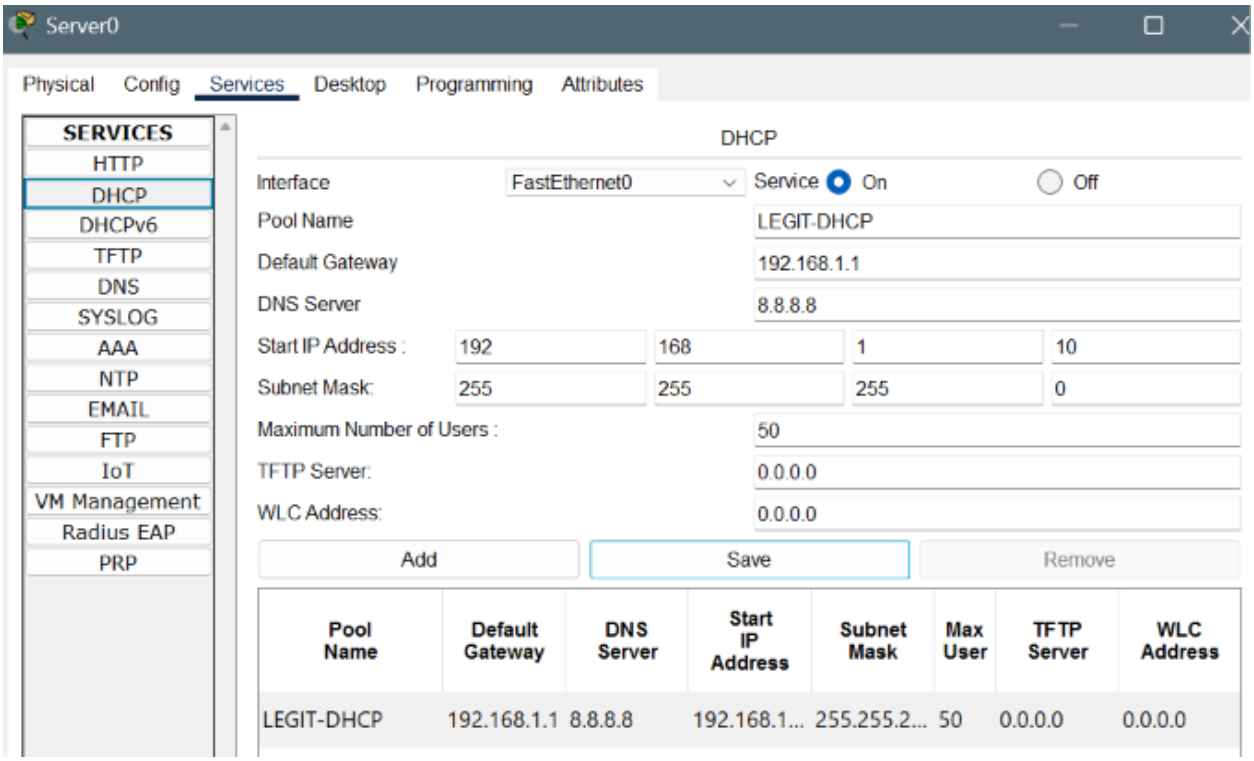
- Detection of rogue DHCP server behavior.
- Configuration of trusted and untrusted interfaces.
- Creation and verification of DHCP Snooping bindings.
- Activation of DAI on VLAN 1.
- Source MAC, destination MAC, and IP address validation.
- Successful legitimate ARP communication.
- ARP packet observation using Packet Tracer Simulation Mode.

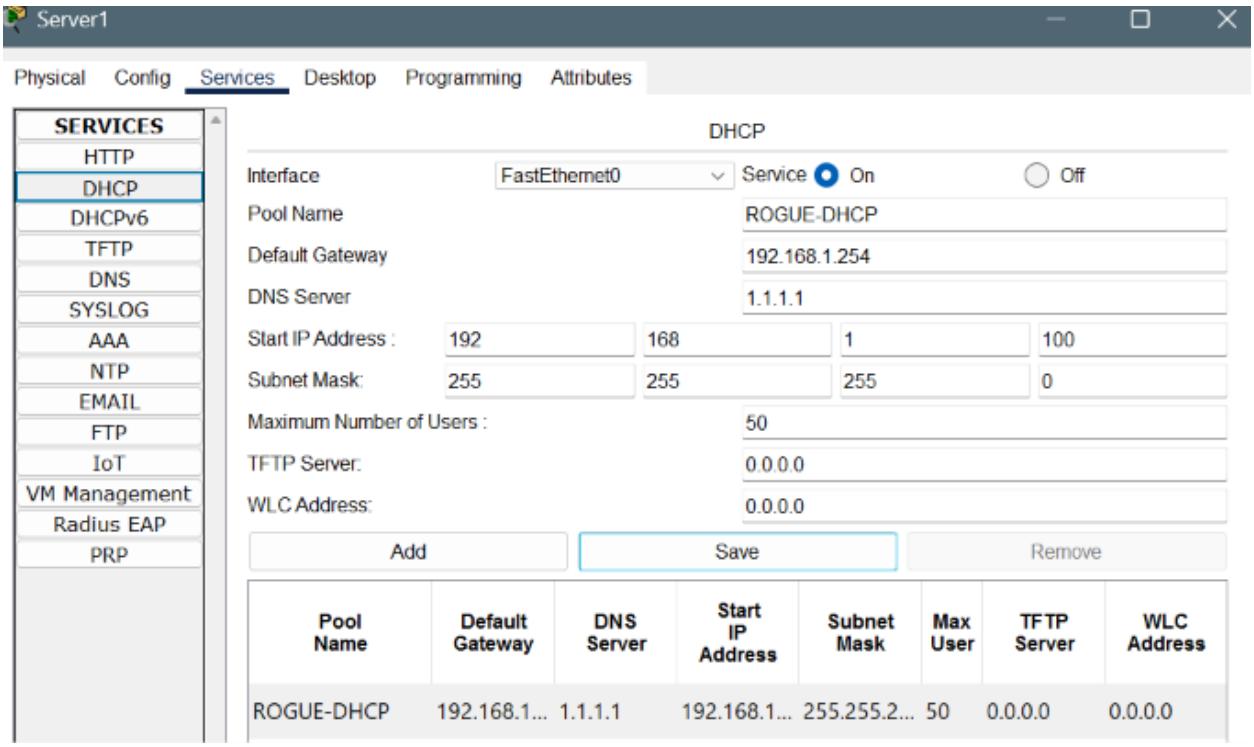
Skills Practiced: Cisco Switching, Layer 2 Security, DHCP Snooping, Dynamic ARP Inspection, ARP, MAC Address Validation, Network Troubleshooting.




PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.14
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
8.8.8.8


Server-PT
Fa0
Fa0
Server-PT
Server1
Fa0/1
Fa0/2
Fa0/3
Fa0/4
Fa0/5
Fa0
PC-PT
PC0
Fa0
Fa0
PC-PT
PC1
Fa0
PC-PT
PC2


Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
LEGIT-DHCP
Default Gateway
192.168.1.1
DNS Server
8.8.8.8
Start IP Address :
192
168
1
10
Subnet Mask:
255
255
255
0
Maximum Number of Users :
50
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
LEGIT-DHCP
192.168.1.1
8.8.8.8
192.168.1...
255.255.2...
50
0.0.0.0
0.0.0.0


Server1
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
ROGUE-DHCP
Default Gateway
192.168.1.254
DNS Server
1.1.1.1
Start IP Address :
192
168
1
100
Subnet Mask:
255
255
255
0
Maximum Number of Users :
50
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
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Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
ROGUE-DHCP
192.168.1...
1.1.1.1
192.168.1...
255.255.2...
50
0.0.0.0
0.0.0.0